Chapter 1 lecture script: Ecological models and data

Draft for the course author's review. **Target: 60 minutes**: 57 minutes on 19 slides, including the questions, figure discussions and short activities below, and 3 minutes of slack for discussion. Course logistics sit outside this hour. The remaining hour of a two-hour session is available for your logistics, a break and further questions; no particular allocation of that remaining time is assumed.

This script follows Chapter 1 theory, with a short introduction to the R practical. It has the same slide text and speaker notes as `lecture.qmd`. The slides carry keywords only; the narrative is in the notes. The notes are a speaking guide: the time allocations include pointing to figures, taking answers and discussing them.

Your two existing figures are used on slides 7 (`ModelCycle.png`) and 14 (`Estimation.png`). Slides 9, 10 and 12 use PNG copies of figures of the chapter, kept in `lecture-assets/`, and slides 5 and 6 use the schematics `feeding-response.svg` and `model-components.svg`. No fitting code is needed for this lecture.

Timing and slide sequence

Times are measured from the start of the Chapter 1 presentation, independently of where you place the logistics in the session.

Slide	Topic	Minutes	Elapsed time
1	Ecological models and data	1	0–1
2	Overview of the lecture	1	1–2
3	The problem and the purpose of the course	2	2–4
4	Ecological questions	3	4–7
5	A predator's feeding response	4	7–11
6	The components of a statistical model	3	11–14
7	The modelling cycle	4	14–18
8	Observation error and process variation	4	18–22
9	Probability models for ecological observations	3	22–25
10	Explaining some of the variation	4	25–29
11	Parameters and estimates	3	29–32
12	Parameter uncertainty	4	32–36
13	Discussion: which source of variation?	3	36–39

Slide	Topic	Minutes	Elapsed time
14	Reality, model and data	4	39–43
15	What we want from a statistical analysis	4	43–47
16	Course roadmap: distributions to estimation	2	47–49
17	Course roadmap: response curves to optimisation	2	49–51
18	The Chapter 1 R practical	3	51–54
19	Summary and closing questions	3	54–57
	Slack for discussion	3	57–60
	Total	**60**	

Slide 1 — Ecological models and data

The lecture has five parts.

First, the kind of question that the course wants to answer and what a model is (slides 3 to 6). We use one example throughout, the feeding rate of a predator.

Second, the modeling cycle, how do we organize our work (slide 7 & 15).

Third, why observations vary and why we describe that variation with probability (slides 8 to 10).

Fourth, what the parameters of a model are and why the values that we calculate from data are uncertain (slides 11 to 14).

Fifth, the chapters of the course and the R practical of this week (slides 16 to 19).

Today is an overview. You do not need to know how to fit any of the models that appear in the examples.

Slide 3 — The problem and the purpose of the course

The purpose of this course is to learn how to use ecological models and statistics to answer ecological questions with data.

We learn models in ecology courses: functional responses, growth curves, responses to the environment. They often stay theoretical. In experimental research statistics dominates, and the statistics that we learn first was developed for factorial experiments, linear responses and Normal distributions. Ecological variables do not respond linearly to their drivers, they do not follow Normal distributions, and a lot of our data do not come from a balanced factorial experiment, because things go wrong in the field or the design is not logistically possible.

The proposal of the course is to combine both disciplines. We take the response functions of ecological modelling and the distributions that describe how ecological variables behave, and we combine them with the same statistical machinery that is behind the linear and generalised linear models.

The course has two objectives. The first is to teach likelihood-based inference, in theory and in implementation, so that you understand better what tools like `glmmTMB` do with your data. The second is to teach how to build non-linear (mixed) models and how to apply to them the same type of analysis.

Delivery

Optional personal introduction: the chapter contains your account of reading Bolker's *Ecological Models and Data in R*. If you want to include it, substitute that account for part of these notes and keep the slide within two minutes.

Slide 4 — Ecological questions

The way we ask a question decides what we need from an analysis. "Does temperature affect development?" has a yes or no answer. "How does development change across the observed temperature range?" names the relationship that we have to describe, and asking for an optimum gives a reason to consider a curve that rises and then falls.

In the same way, a treatment could change seedling survival by a small or a large amount, with different consequences for recruitment. We need the size of the change and its uncertainty.

So in this course we give up null hypothesis testing as the main goal of an analysis, because the questions that we care about are quantitative. This does not mean abandoning linear models; slide 15 comes back to it.

A useful habit is to finish this sentence: "To answer my ecological question, I need to estimate…" That might be a probability of survival, a maximum feeding rate or a difference between groups.

Short activity — included in the three minutes

Give students 30 seconds to suggest a more specific version of "Does food availability affect growth?" Take one or two answers. Possible answers: the increase in growth at low food availability, the maximum growth rate, or the food availability needed to reach a given growth rate. Accept answers that identify a quantity and a range or comparison.

Transition

Let us use a feeding response to see how such a question guides a model.

Slide 5 — A predator's feeding response

Imagine that we measure the feeding rate of a predator across a range of prey densities. We could ask whether prey density has a statistically significant effect. More useful questions are: how quickly does feeding increase when prey are scarce? Does it reach a maximum? What is that maximum, and how uncertain is the estimate?

A straight line might describe part of the observed range, but it cannot keep increasing forever. A predator has limited time to search for, capture and handle prey. A saturating response curve represents that constraint directly, and its parameters describe features of the system that we can

discuss and compare among studies. Those comparisons only make sense when the model definitions, units, organisms and study conditions are similar enough.

We often know something about these shapes from theory, earlier experiments or from thinking about how organisms work. Convenience used to be a good reason to fit a straight line anyway, because more complicated models were difficult to fit. Modern computers remove that limit. The challenge now is whether the model represents the ecological relationship reasonably well and whether the data can inform its parameters.

Comparing the values of parameteres across sites, species, treatments or other factors is one way to answer ecological questions. Sometimes we will also compare different models that are derived from alternative hypothesis. This is how we do statistics when we do not have yes/no questions.

Slide 6 — The components of a statistical model

A statistical model is a mathematical approximation to the process that produces our data. That process includes what happens ecologically and how we observe it.

In the feeding example, one part of the model describes how expected feeding changes with prey density. This is a deterministic function, and we use deterministic functions to approximate the ecological process that we are most interested in. The other part describes the possible observations at each prey density: different predators eat different numbers of prey under the same conditions.

A statistical model needs at least one stochastic component, that is, a distribution. The response function is the part that we add when the question asks for it. Some questions need only distributions, for example the counts under one set of conditions, and a model can even be made of nested distributions. The course starts with those models and adds response functions and groups later.

Delivery

The figure is the curve of slide 5 with hypothetical observations added: several predators at each of seven prey densities. Ask students which component describes the curve and which describes the scatter. Point to one prey density and note that the predators differ although the conditions are the same. Keep this conceptual; distribution names and equations come later.

Slide 7 — The modelling cycle

This figure is the map of the course. Ecological modelling begins with a research question, but it does not move in a straight line from question to answer. Each result may send us back to the question, the data or the model.

The three boxes on the right are the three parts of the work, with the chapters where we learn each tool.

Model: we express an ecological idea as a model that can generate the kind of data that we observe. We choose a probability distribution (Chapter 2), describe how its parameters change with continuous predictors (Chapters 5 and 6) and how they vary across groups such as location, species or treatment (Chapter 7).

Estimation: given a model and data, we estimate the parameter values that make the model most compatible with the observations. Chapter 3 introduces estimates, their sampling variation and confidence intervals. Chapter 4 develops maximum likelihood, Chapter 8 deals with the technical difficulties of finding the optimum, and Chapter 7 covers the marginal likelihood that hierarchical models need.

Model comparison and answer: more than one model may be plausible. Chapter 4 introduces the Akaike information criterion, AIC, to compare fitted candidate models. Chapter 6 adds another information criterion, checks how well the models describe the data and interprets the parameters in terms of the question. We compare fitted candidates; we do not assign probabilities to models.

Estimation has not been explained yet. Every box gets its own slide in the next half hour, and the figure returns in the summary.

Figure walkthrough — included in the four minutes

Trace question, model, data, estimation and answer with a pointer, then the loop through model comparison. Ask: "Suppose our fitted curve still rises at the highest prey density that we measured. What might we revisit?" Accept extending the measurements, reconsidering the candidate functions, or restricting the claim to the observed range.

Slide 8 — Observation error and process variation

We use probability as a tool to describe unexplained variation and the uncertainty that follows from it. There are three sources to keep apart: observation error, process variation and parameter uncertainty. This slide defines the first two; slides 11 and 12 take up the third.

Observation error is the difference between what happens and what we record. We may miss an individual or count it twice, instruments never give perfect readings, and protocols are not followed perfectly. In the hypothetical case, twenty insects are on the plant and we record sixteen. Observation error changes the data that we analyse and leaves the ecological reality as it was. How the observations are collected matters for the probability model, and repeated observations under the same conditions help to characterise the error.

Process variation, also called process error or stochasticity, is variation in the ecological reality that the deterministic part of the model does not explain. It is still there with perfect measurements. "Error" does not mean that somebody made a mistake. The slide lists four reasons: the environment and drivers that we did not measure, differences among individuals, chance in births and deaths, and nearby or repeated observations that resemble each other. The chapter adds the non-linear effects and interactions that a model simplifies away.

Unexplained is not the same as random. Persi Diaconis of Stanford University built a model that predicts coin flips, provided that the initial conditions are measured accurately enough. Process variation is therefore relative to the model and to the information that we have.

A recorded value often reflects both sources. Simple models describe their combined variation with one distribution. When the data have groups, hierarchical models (also called mixed or mul-

tilevel models) can separate them, or at least estimate a good part of the process variation. That is Chapter 7.

Discussion

Ask the question on the slide and take one reason for the answer. Relate it to one of the four items on the right.

Slide 9 — Probability models for ecological observations

A probability distribution describes the possible values of an observation and how probability is allocated to them.

The two panels show why we need several distributions. Insect counts are whole numbers that start at zero. Body mass is a positive continuous quantity, and here it is strongly skewed: many small values and a few much larger ones. The grey bars summarise the observations and the orange marks or curve show a fitted model, a Negative Binomial distribution on the left and a LogNormal on the right. We will seldom use the Normal distribution in this course.

The vertical axes differ. For counts we assign a probability to each possible count. For a continuous measurement, probability is an area under a density curve over an interval. Chapter 2 develops that distinction.

Figure discussion — included in the three minutes

Ask students to identify the observations and the model in each panel. Then ask: "Could an insect count be negative? Could it be 3.5?" A model should respect the possible values of the quantity that it describes. Do not start a detailed comparison of distributions.

Slide 10 — Explaining some of the variation

These are the insect counts of the previous slide. The plot now shows what the pooled plot hid: the plants were sprayed with different products.

There is variation among the sprays and within each spray. The variation among sprays is process variation, because we do not expect our counts to have errors that depend on the spray: the sprays changed how many insects were on the plants.

"Explained" and "unexplained" always refer to a model. For the pooled model of the previous slide, the differences among sprays are unexplained process variation. For a model that includes spray, they are explained. The variation within each spray remains, and it contains process variation among plants and observation error. The figure cannot tell us how much there is of each.

This is how the models of the course grow: we describe how a response changes with a predictor (Chapter 6) or how parameters vary among groups (Chapter 7), and we keep a probability distribution for the variation that remains.

Figure discussion — allow about two minutes

Ask the question on the slide. If there is time, ask: “What variation remains after we account for spray?” Point to the differences among groups and to the spread within each group. Describe what is visible without making a significance claim.

Transition

Once we propose a model, we still have to determine its parameter values.

Slide 11 — Parameters and estimates

A parameter is a quantity that determines a feature of the model. A model for survival has a parameter for the probability of survival. A feeding response has a parameter for its maximum rate. A distribution also has parameters that describe how much the observations vary.

The parameter belongs to the model. The estimate is the value that we obtain when we apply an estimation procedure to the observations, and a different procedure can give a different value. To estimate, we treat the data as if they were a random sample from the model; slide 14 looks at that assumption.

Another sample from the same study design would give a different estimate. This is the third source, parameter uncertainty, and it is the reason why we report more than a fitted value.

Chapter 3 introduces this reasoning with repeated hypothetical samples. Chapter 4 develops maximum likelihood, the estimation method of the rest of the course.

Check for understanding

Point back to the feeding curve: “Which feature could be controlled by a parameter?” Take one answer, then ask what data would help to estimate it.

Slide 12 — Parameter uncertainty

The same model is compatible with different samples of data, because of the variation that the model does not explain. Turn this around: with the data fixed, several models are compatible with that sample, and each corresponds to different values of the parameters. That we do not know exactly which values to use is what we call uncertainty.

The points are the body mass and brain mass of 28 species of land animals, on logarithmic axes because the sizes cover a wide range. The line is a fitted saturating response and the band is an approximate 95% confidence interval around the fitted mean response. The band is not where we expect individual species to lie. It shows that several similar curves are compatible with these data.

Three of the 28 species are dinosaurs (*Diplodocus*, *Triceratops* and *Brachiosaurus*) with very large bodies and small brains. The levelling-off of the curve at large body masses comes from them.

There are two formal definitions of parameter uncertainty, the frequentist and the Bayesian. For large samples and simple models they differ little. For small data sets the Bayesian way can be superior, but it is sensitive to the prior assumptions that complement the data. We keep to the frequentist interpretation, with two things to remember: our statements of uncertainty are ap-

proximate when we have little data, and Bayesian methods are rising in popularity. The book is meant as a bridge to them.

Figure discussion — allow about two minutes

Ask: "What does a point represent? What does the line represent? What does the band represent?" Give students time to separate the three. Keep to interpretation; the calculation of confidence intervals comes in Chapters 3 and 4.

Slide 13 — Discussion: which source of variation?

Case 1: the animals are the same and the records differ. This is observation error.

Case 2: say aloud that the model has only spray as a predictor; the slide leaves it out. The counts are the true ones, so nothing was recorded wrongly. Relative to that model the difference is process variation. It would become explained variation if we measured, say, plant size and added it to the model.

Case 3: what changes is the estimate from one sample to another. This is parameter uncertainty. Each sample may contain many observations; the variation here is among the estimates of hypothetical samples.

All three occur in one analysis. We separate them to be clear about what a distribution or a statement of uncertainty describes.

Slide 14 — Reality, model and data

This figure zooms into the estimation box of the cycle. With the cycle on slide 7, it is one of the two figures of the chapter.

The data generating process is everything in a study that can affect the data that we collect: the ecological processes, the experimental protocols and the measurement procedures. For the feeding experiment it includes encounters with prey, capture, handling, differences among predators and how feeding is recorded. The model is a mathematical approximation to that process, and its parameters give it the flexibility to match different possible processes.

For the purpose of estimation we treat the data as if they were a random sample from the model. In reality they come from the data generating process and not from the model. We need the assumption to develop the statistical procedures: it links the parameter values to the data sets that we could have observed.

A parameter can be estimated with many procedures. Frequentist statistics studies how a procedure behaves across repeated hypothetical samples: whether the estimates are too high or too low on average (bias), how much they vary, and how often their intervals cover the parameter. Chapters 3 and 4 develop this.

We use maximum likelihood, the most general procedure. Informally, it selects the parameter values that make the model most compatible with the observed data. Chapter 4 defines it.

Figure walkthrough

Follow the solid arrows, then the dashed arrows, with the labels of the figure. Ask where a missed insect or an imperfect instrument belongs in the diagram: in the data generating process, which includes how we observe.

Slide 15 — What we want from a statistical analysis

Most of us first learn statistics through linear models and ANOVA, and later perhaps generalised linear and mixed models. They are useful tools and you should not throw them away. But they are often taught as if the challenge were to choose the right formula, check a few assumptions and see whether a coefficient differs from zero. That is not always the question that we care about.

A small p-value is rarely the end of an ecological analysis, or its main goal. A test against a zero effect does not tell us whether the effect is large enough to matter, what shape the response has or how precisely we know it. "Nothing happens" is often not even a plausible ecological model: we usually choose among several ways in which something might happen. And a large p-value often says more about the size of the experiment or survey than about the effect. For the debate, the chapter recommends Wasserstein, Schirm and Lazar (2019), "Moving to a world beyond 'p < 0.05'", *The American Statistician* 73(sup1), 1–19.

The approach of the course is to make the choices explicit: a response function, a probability model for the observations, estimates with uncertainty, a comparison of plausible candidates and a check of the fit. The aim is the simplest model that answers the question. If that is a linear model with Normal errors, we use it, because we choose to and not because we have to.

Two limits of scope. Probability also describes uncertainty about initial conditions, inputs such as future rainfall, scenarios and numerical error. These matter for dynamical and forecasting models, which the course does not cover. And we compare candidate models without assigning probabilities to them.

Stating the model, estimating with uncertainty and checking the fit are also useful habits when models inform decisions, although this course is about analysing data and not about forecasting.

Discussion

Ask: "If I tell you that prey density has a statistically significant effect, which of our questions about the feeding curve are still unanswered?" Refer to slide 5 and take two answers.

Slide 16 — Course roadmap: distributions to estimation

One sentence per row.

Chapter 2: what values an observation can take and with what probabilities, with a Binomial model for seed survival and a Normal model for penguin flipper lengths.

Chapter 3 changes the direction of the question: we have a sample and want to learn about unknown parameters, so we study how samples and estimates vary.

Chapter 4: maximum likelihood, with the uncertainty of the estimates and a first comparison of models with AIC.

The R tools arrive when we need them: the distribution functions of the packages RTMB and RTMBdist from Chapter 2, and fitting with RTMB and `optim()` from Chapter 4.

Delivery

Stress the change from Chapter 2 (known parameters, possible observations) to Chapters 3 and 4 (observed data, unknown parameters). Do not list every distribution or method.

Slide 17 — Course roadmap: response curves to optimisation

One sentence per row.

Chapter 5: the response functions themselves, their shapes and the meaning and units of their parameters, without fitting.

Chapter 6: functions, distributions and maximum likelihood together, in a complete analysis that fits, compares, checks and interprets.

Chapter 7: groups and repeated observations, with hierarchical models.

Chapter 8: how the numerical searches work and what to do when a fit is difficult.

The supplement on formula-based methods describes `glmmTMB`, which uses the same techniques as the course, with a formula that writes the likelihood for you. This closes the loop with the first objective of slide 3.

The detailed roadmap in the chapter is a reference to return to while you work through the course.

Slide 18 — The Chapter 1 R practical

The Chapter 1 practical teaches the R skills that the rest of the course uses. R is the language for the calculations and RStudio helps us to write and organise the work.

We store commands in scripts so that an analysis can be run again. We learn how to store observations, select and summarise data and make plots. Functions let us write a model or a calculation once and use it for different inputs. Loops repeat operations, which we need when we simulate studies later in the course.

Work through the tutorial in order and run the examples yourself. When you change a line, predict what will happen before you run it.

The practical also covers the installation of packages, finding help and the use of AI. You may use AI as a tutor: to translate the material, explain concepts, provide examples and fix bugs in your code. If it writes the solutions before you have tried, you will not learn the material. AI tools make mistakes and you are responsible for your work, so you must be able to explain what your code does and what its result means ecologically.

Delivery

Open the practical briefly, or show a screenshot of its contents, and point out the sections on scripts, functions and exercises. Keep installation, schedules and assessment for the logistics segment. A live demonstration is optional and replaces part of the three minutes of this slide.

Slide 19 — Summary and closing questions

Go back to slide 7 and show the cycle again; no new slide is needed. We started from a question, expressed an ecological idea as a response function with a probability distribution, and saw what estimation assumes and why estimates are uncertain. Comparison and checking send us back around the cycle until the answer is useful for the question.

An estimate is useful when we know what it measures, its units and its uncertainty.

Return to this slide for the two closing questions.

Discussion and expected answers — allow about two minutes

Question 1: the curve describes the expected response, and observations at the same prey density differ. A probability model represents that variation, and estimation and uncertainty are built on it.

Question 2: variation among observations is in the data and contains process variation and observation error. Parameter uncertainty is about how precisely a sample determines a quantity of the model; repeated hypothetical samples give different estimates.

Final transition

The next steps are the R tools of the practical and the probability language of Chapter 2.

Adjustments during delivery

The slides take 57 minutes and leave 3 minutes of slack for discussion. If discussion runs longer, use the slack first and then shorten slides 16 and 17 to one spoken sentence each, which recovers about three minutes. Keep the distinction between variation and parameter uncertainty and the explanation of the two figures (slides 7 and 14).

If you prefer a break within the Chapter 1 content, place it after slide 13 (39 minutes of content). The break is additional to the hour.

If time remains in the wider session, return to the feeding example and ask students to propose an experiment: what they would measure, which prey densities they would include and which observation problems they expect.